



Time Series
Analysis

Time Series Analysis

Today's Topic: Financial Time Series II



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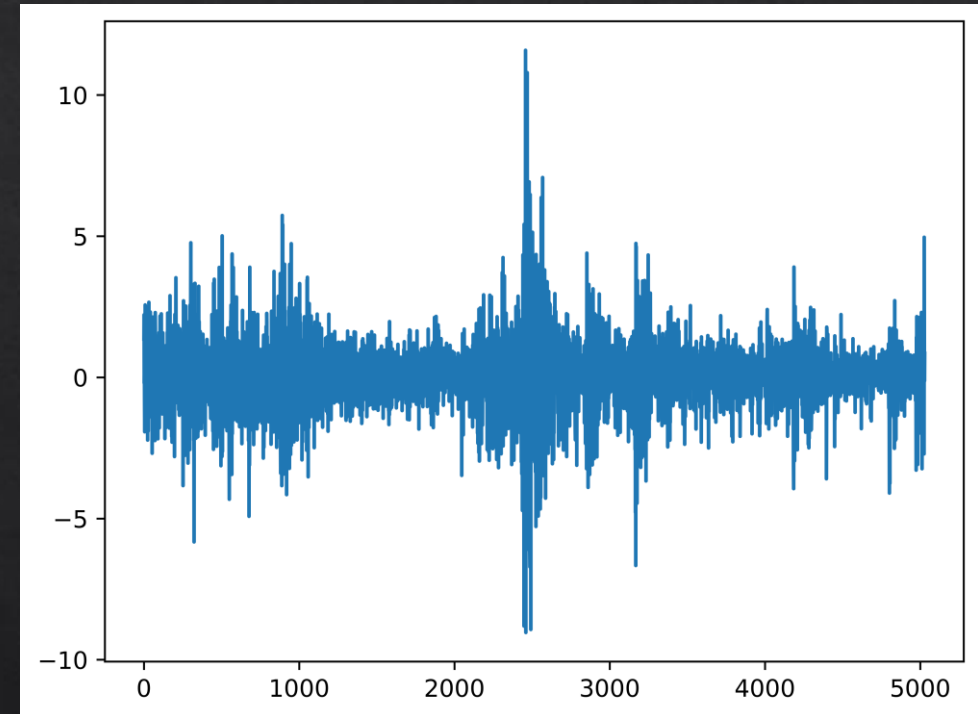


<https://sithuaung-ink.github.io/data-analyst-portfolio/>

ARMA Model

Standard & Poor's 500 Index Daily Index

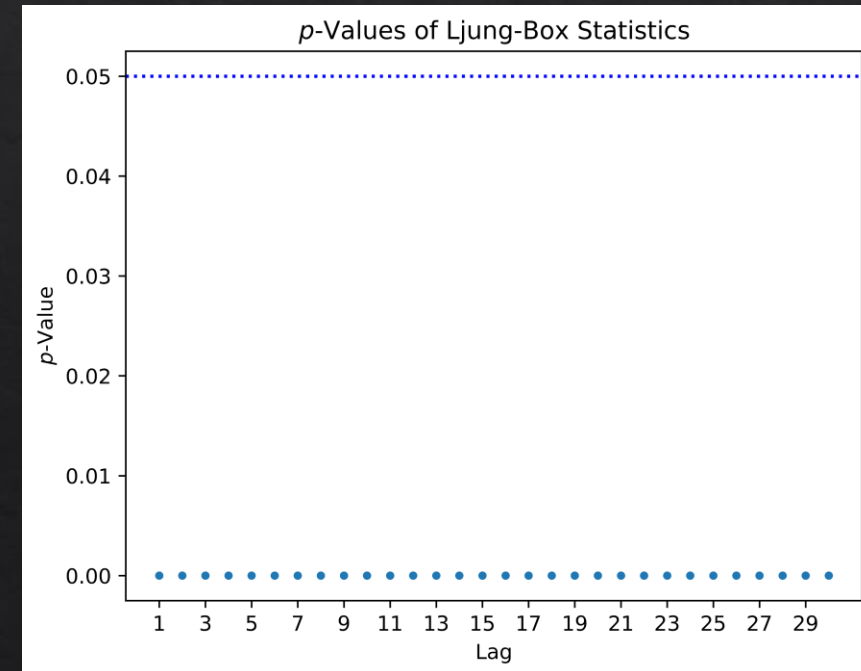
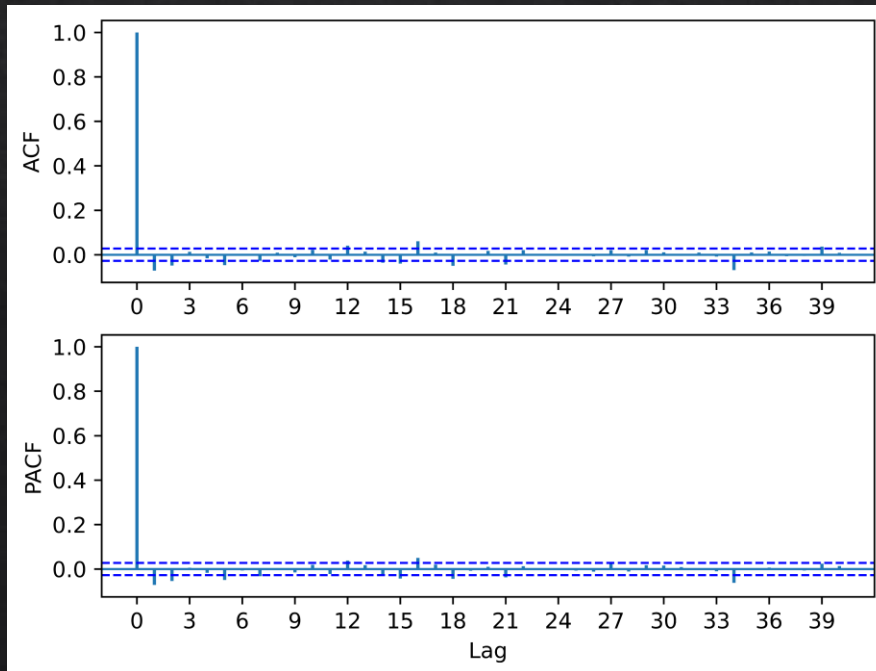
- SP500 index daily returns from Jan 5, 1999, to Dec 12, 2018.
- The time series plot is shown in the figure.



ARMA Model

Standard & Poor's 500 Index Daily Index

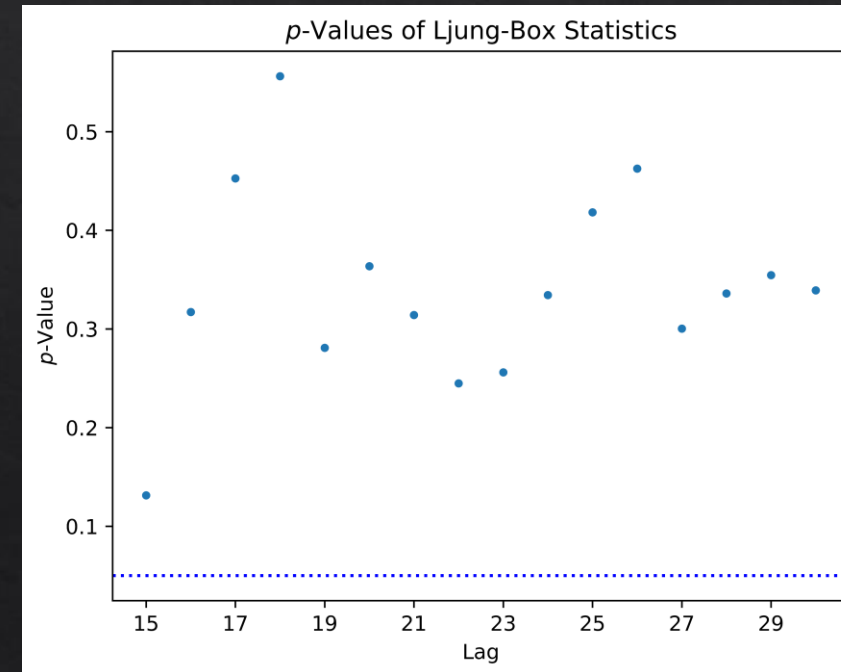
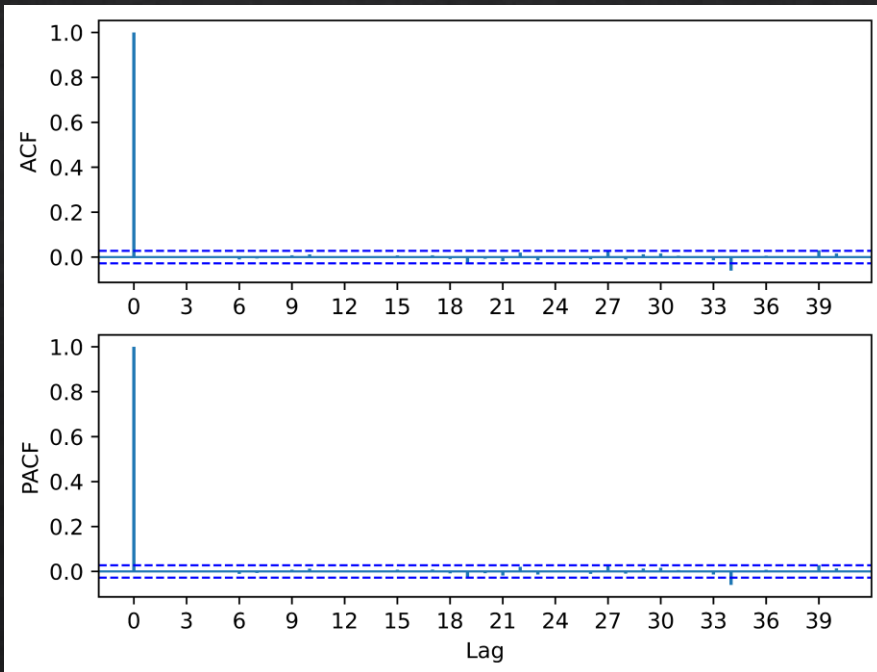
- The return series is stationary (KPSS Test).
- The ACF and PACF plots and the p-value plot shown in the figure show that the series is not white noise.



ARMA Model

Standard & Poor's 500 Index Daily Index

- Build an adequate ARMA model for the series.
- Estimated ARMA (15, 2) model is obtained after dozens of trials¹.
- Refined this model by fixing the parameters.

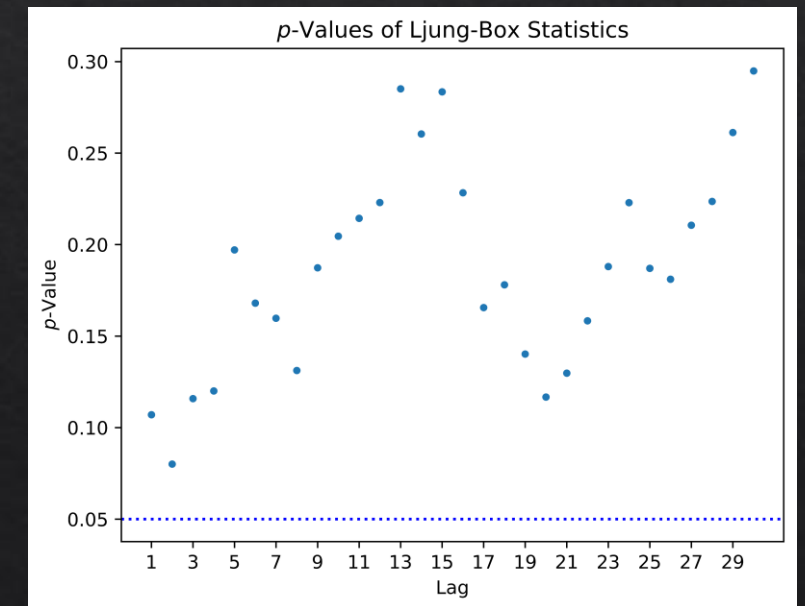
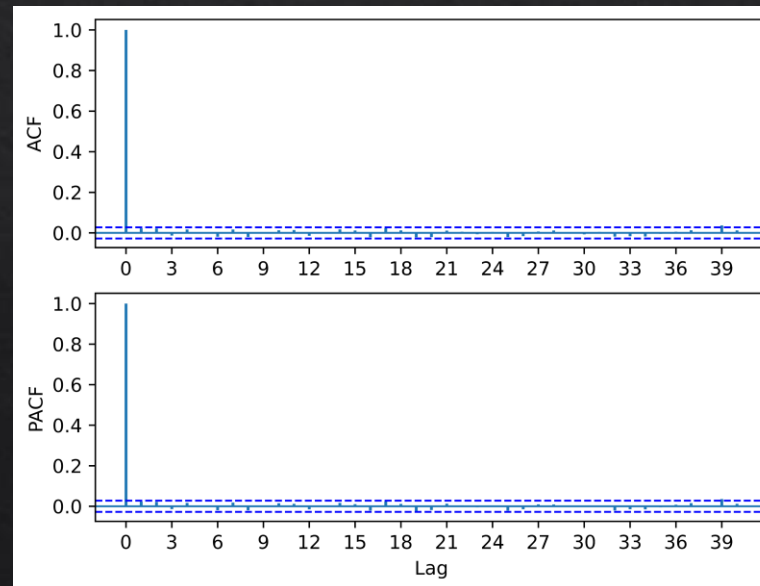
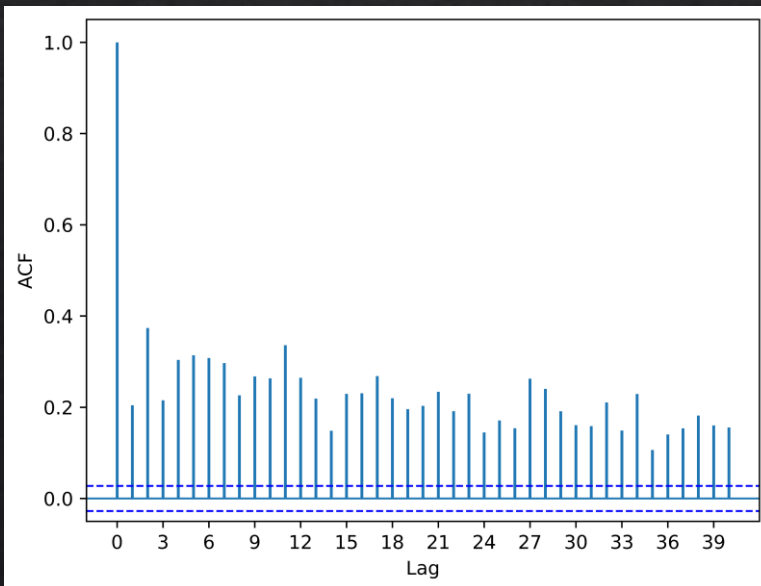


¹Huang, C., & Petukhina, A. (2022). Applied time series analysis and forecasting with Python (Vol. 39). Cham: Springer.

GARCH Model

Standard & Poor's 500 Index Daily Index

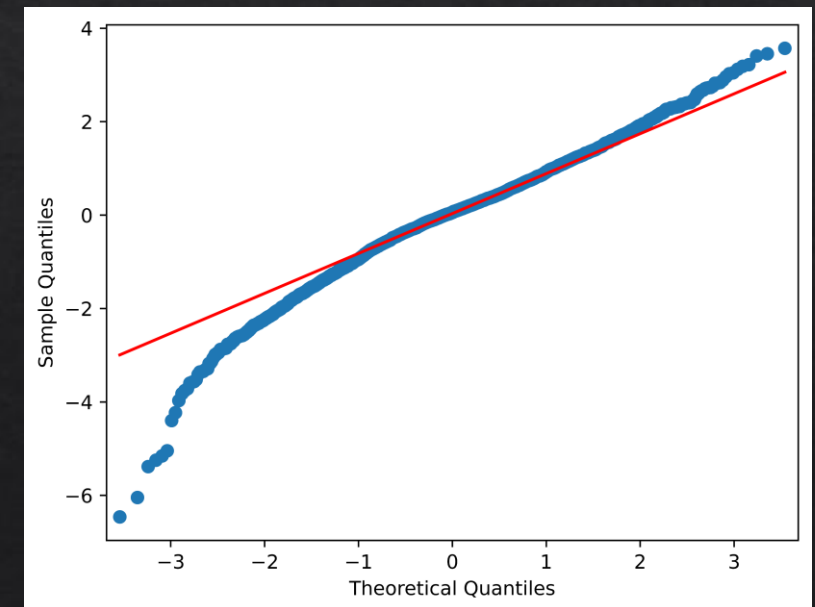
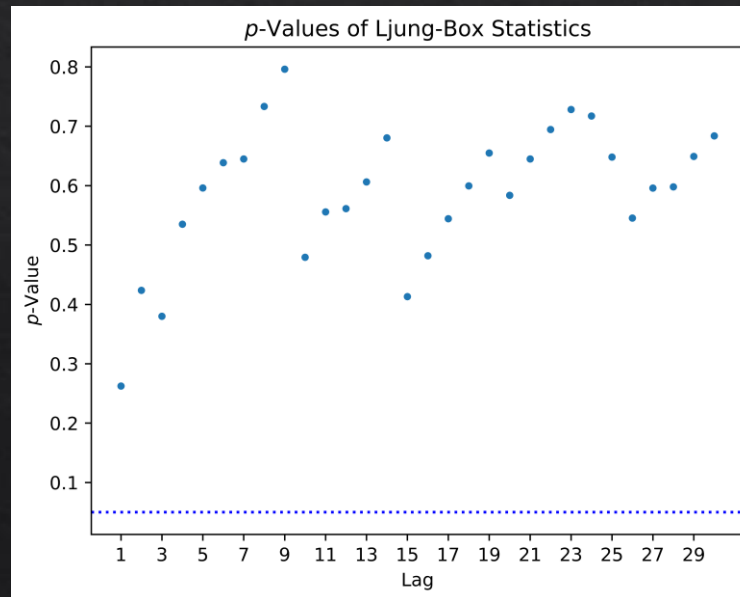
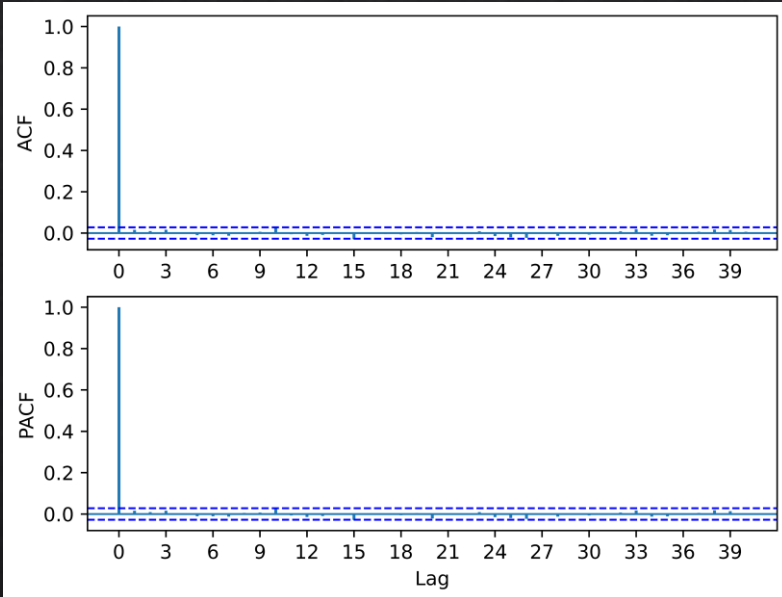
- The residual series has an ARCH effect.
- Build an adequate GARCH model for the return series using the mean model residuals.
- GARCH (2, 2) models will be selected, and the model residuals are uncorrelated and have no ARCH effect anymore.



GARCH Model

Standard & Poor's 500 Index Daily Index

- GARCH model squared residuals are uncorrelated and have no ARCH effect anymore.
- The QQ plot indicated that the left tail of the empirical distribution of the residuals is heavier than that of a normal distribution.



- The GARCH model successfully captures a few important features of financial time series.
- However, it has no ability to characterize all the stylized facts on financial time series.
- The financial market reacts differently to bad news than to good news.
- There is a certain amount of asymmetry that cannot be caught by the GARCH model.

EGARCH Model and TGARCH Model

An EGARCH(p, o, q) (exponential GARCH) model

$$\begin{cases} X_t = \sigma_t \varepsilon_t, \varepsilon_t \sim iid N(0,1). \\ \ln \sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i (|\varepsilon_{t-i}| - E|\varepsilon_{t-i}|) + \sum_{j=1}^o \gamma_j \varepsilon_{t-j} + \sum_{k=1}^q \beta_k \ln \sigma_{t-k}^2 \end{cases}$$

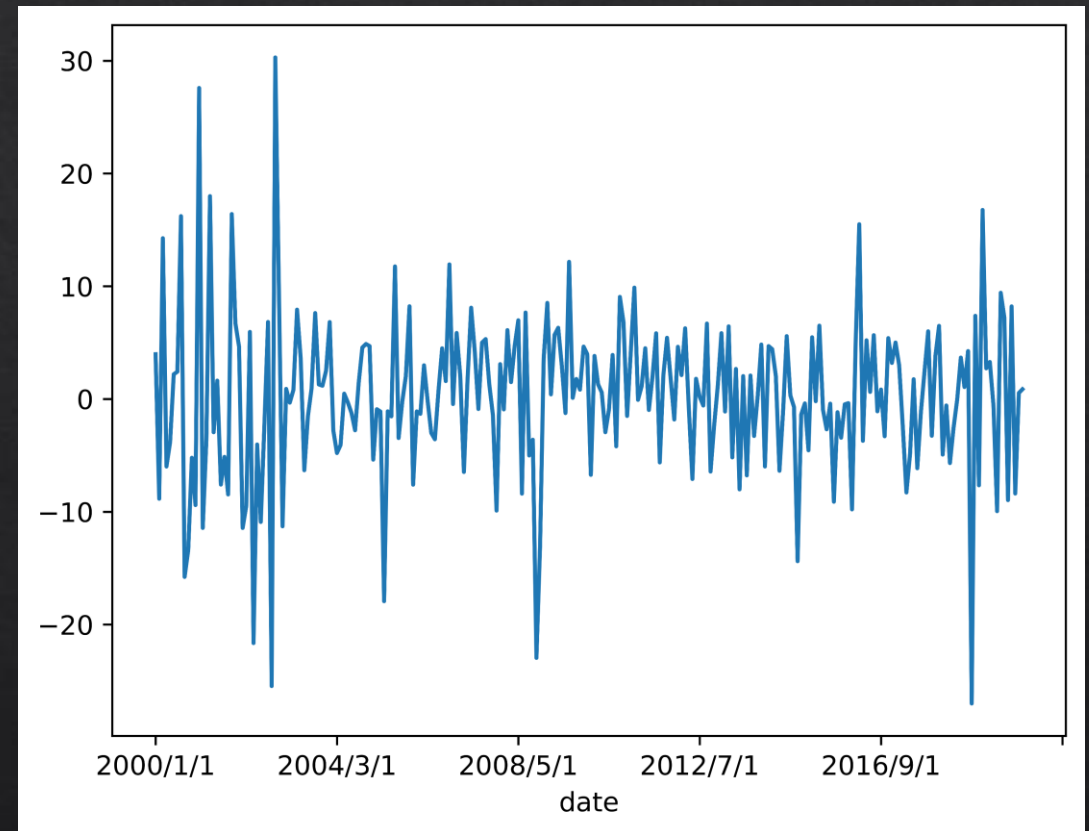
An TGARCH(p, o, s, q) (threshold GARCH) model

$$\begin{cases} X_t = \sigma_t \varepsilon_t, \varepsilon_t \sim iid(0,1). \\ \sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i |X_{t-i}|^s + \sum_{j=1}^o \gamma_j |X_{t-j}|^s I_{[X_{t-j} < 0]} + \sum_{k=1}^q \beta_k \sigma_{t-k}^s \end{cases}$$

ARMA Model

IBM Stock Log Returns

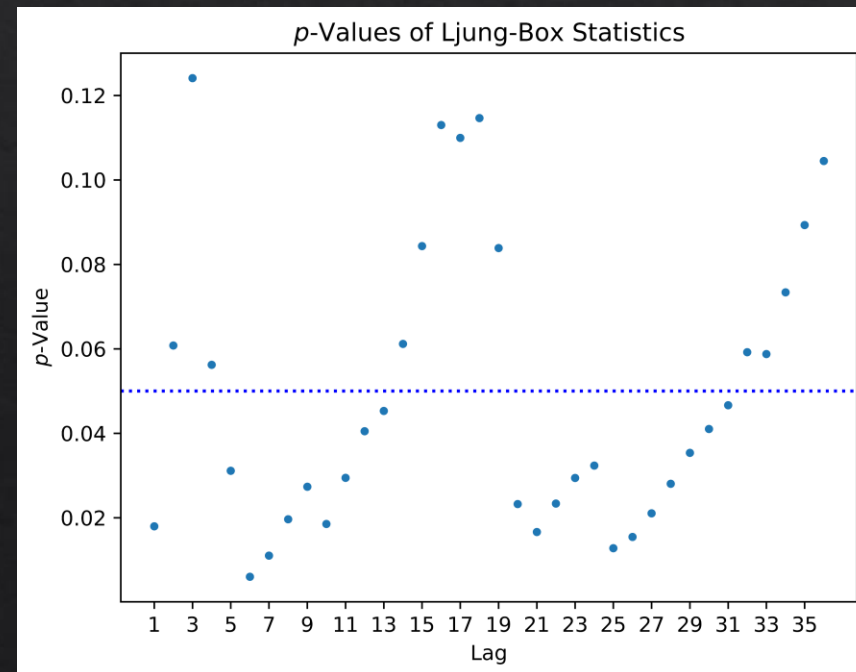
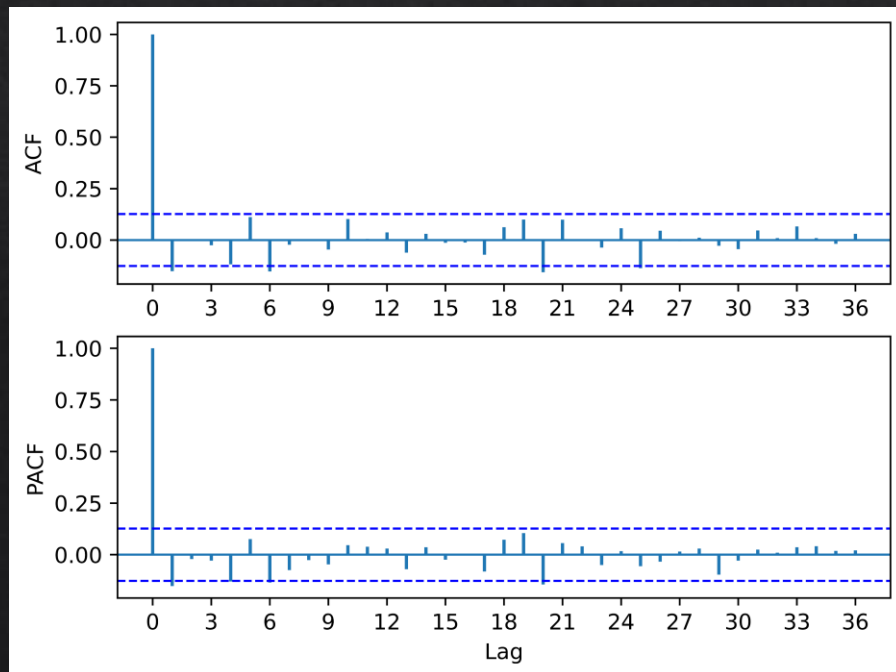
- IBM stock from January 2000 to December 2019.
- The time series plot is shown in the figure.



ARMA Model

IBM Stock Log Returns

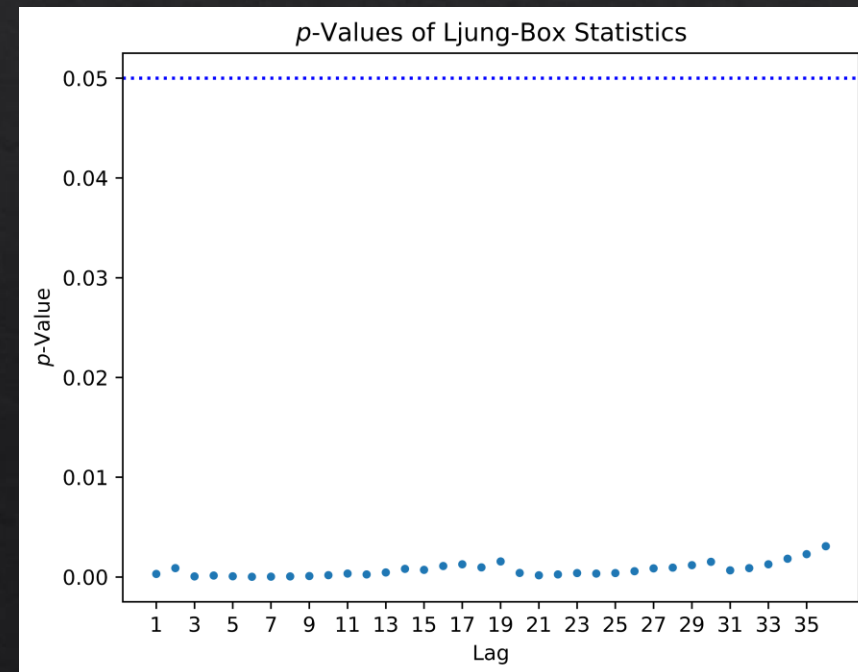
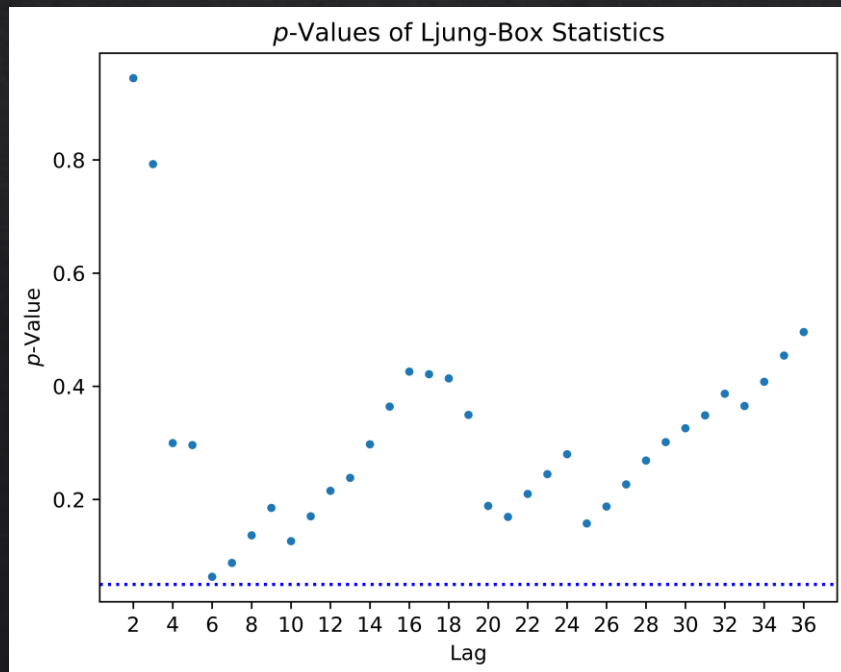
- The return series is stationary (KPSS Test).
- The ACF and PACF plots and the p-value plot shown in the figure show that the series is not white noise.



ARMA Model

IBM Stock Log Returns

- Build an adequate ARMA model for the series.
- An estimated ARMA (0, 1) model is selected, and the residuals have the ARCH effect.
- Try to build an appropriate GARCH model for this series, and it turns out there is no adequate GARCH model for it.



EGARCH Model

IBM Stock Log Returns

- The EARCH model is decided to use.
- The p-value plot for the Ljung-Box test of the residuals of the estimated EGARCH model is uncorrelated, and the p-value plot for the Ljung-Box test of the squared residuals has no ARCH effect anymore.
- The QQ plot of the residuals is satisfactory except for the left tail.

